Hildebrand: corrected box asymptotic proved

Result. Lean proves the corrected author-preprint Lemma2 for k=d+1≥3. For every fixed d≥2 and ε>0, there exists a threshold such that every sufficiently large prime has actual uniform conditional box probability ≤1.1·(ε/2)/(2^d·p). The threshold is independent of all integer cell shifts, first support point a_0 and nonzero Fourier index j. [pcc:unverified]

Exact finite core	Actual uniform conditional injective support law Probability ≤ (W+1)^d / (p−1).descFactorial d
Derived asymptotics	d≥2: W→∞ and actual denominator / p^d → 1 Exact W^d = ((ε/2)/2^d)·p^(d−1); correction → 1
Corrected result	Uniform in cell shifts, a_0 and nonzero index j For all sufficiently large primes: probability ≤ 1.1·(ε/2)/(2^d·p)

Actual source and literal formulas. The sampler covers all injective (d+1)-point field supports with fixed first point. Its actual cardinality is (p−1).descFactorial d; its mapped PMF equals the uniform law on the full conditional space. Source coordinates use the actual canonical centered representative of $(a_0-a_i)\cdot j$. Lean binds explicit identities with the printed real-power endpoints $m_i\cdot(\varepsilon/2)^{(1/d)}\cdot p^{((d-1)/d)}/(2p)$ and the corresponding m_i+1 upper endpoint. [pcc:unverified]

Limits are derived. Actual factorial normalization equals the product of factors1−i/p and tends1. The unnormalized width W=(ε/2)^(1/d)·p^((d−1)/d)/2 tends to infinity when d≥2 and satisfies the exact dth-power identity. Hence (1+1/W)^d divided by the normalized denominator tends1. An exact finite-ratio identity and eventual strict bound11/10 yield the printed1.1 constant. No asymptotic, probability, cardinality or mixing premise is supplied. [pcc:unverified]

Known k=2 failure retained. Lean proves W(1,ε,p)=ε/4 and directly negates width divergence. The preceding actual k2 cell counter remains in the main certificate: ε=1/m_2=4/a_1=0/j=1 gives probability1/(p−1)>11/(40p) for every prime p≥3, with direct negation of any eventual prime threshold. The later author preprint acknowledges the older proof failure and treats k=2 separately; no novelty claim is made. Qualified mixing theorems are retained. [pcc:unverified]

Sources and remainder. Author preprint math/0401189v1: selected section4.2/pages13–16/reference14p36 read, visual15–16 only; corrected Lemma2 explicitly requires k≥3. Hildebrand1994: full13-page original read/inspection credited; proof197 rechecked. Official Mathlib source APIs and preceding finite-counting proof read. The stronger expected random-support mixing exponent still needs the spectral argument. [pcc:unverified]

Checks. Whole gate: 21 declarations, zero failures, zero trusted native exceptions. All 97 build/processing/provenance files retained losslessly in 37 objects, five gate files and recursive source/config pins. Lean4.35.0/Mathlib56a00585…, standard foundations/default limits. Signed standing unverified; KB/Conductor fallback recorded. No holes/custom axioms/native evaluation/raised limits/global flags/waivers/native experiment/paid compute/upload. [pcc:unverified]